```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
    int fd = open("combined.txt",
                  O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    dup2(fd, STDERR_FILENO);
    close(fd);

    printf("This is standard output\n");
    fprintf(stderr, "This is standard error\n");

    return 0;
}
```

```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <fcntl.h>

int main() {
    int fd[2];

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    pid_t pid = fork();

    if (pid == 0) {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);

        execlp("ls", "ls", NULL);
        perror("ls");
        _exit(1);
    }

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    close(fd[1]);

    int out = open("pipeline.txt",
                   O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (out < 0) {
        perror("open");
```

```c
    }

    close(fd[1]);

    int out = open("pipeline.txt",
                   O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (out < 0) {
        perror("open");
        close(fd[0]);
        waitpid(pid, NULL, 0);
        return 1;
    }

    dup2(fd[0], STDIN_FILENO);
    dup2(out, STDOUT_FILENO);

    close(fd[0]);
    close(out);

    execlp("wc", "wc", "-l", NULL);
    perror("wc");
    return 1;
}
```

```
sravya@Sravya1955:~/os$ nano q15a.c
sravya@Sravya1955:~/os$ gcc q15a.c -o q15a
sravya@Sravya1955:~/os$ ./q15a
sravya@Sravya1955:~/os$ cat combined.txt
This is standard error
This is standard output
sravya@Sravya1955:~/os$ nano q15b.c
sravya@Sravya1955:~/os$ gcc q15b.c -o q15b
sravya@Sravya1955:~/os$ ./q15b
sravya@Sravya1955:~/os$ cat pipeline.txt
159
sravya@Sravya1955:~/os$ ls | wc -l > pipeline.txt
sravya@Sravya1955:~/os$
```